Student Activity 1: Tables and Counts data

A Course on Data Visualization

Livio Finos, University of Padova

2026-10-06

Contents

A two-by-two table **1**
- Source and variable definitions . . . 1
- Absolute frequencies: how many participants? . . . 2
- Conditional proportions: colour-vision status given sex . . . 4
- Reverse the conditioning: sex given colour-vision status . . . 5
- Compare the questions side by side . . . 8
- Optional: a mosaic plot . . . 10
- Reference and reproducibility . . . 10

Relationship between two count variables **12**
- Proposal 1: count plot . . . 13
- Proposal 2: joint-frequency heatmap . . . 14
- Proposal 3 Count plot with jitter . . . 15
- Side-by-side comparison . . . 18
- Suggested classroom question . . . 19
- Session Info . . . 19

A two-by-two table

Source and variable definitions

(Slightly modified for didactic purposes) Ngente et al. (2021), *Prevalence of Color Vision Anomalies among Dental Professionals*, report a cross-sectional study of 198 dental professionals, including students and faculty. Colour vision was assessed using Ishihara plates. Table 2 reports 5 affected participants among 54 males and 4 among 288 females.

Read the article and Table 2.

We call the outcome **colour-vision deficiency (CVD) detected / not detected**, retaining the study's binary classification. "Not detected" refers to this assessment, not the absence of every possible visual condition. We use **sex (as recorded)** for the reported Male/Female categories; the source labels this variable "gender" and does not establish how sex and gender identity were distinguished. These are sample frequencies, not universal prevalence values.

The original treatment example is replaced by these observed counts. No individual-level records are generated or implied.

```
tab <- matrix(c(5, 49,
                4, 284), nrow = 2, byrow = TRUE)
dimnames(tab) <- list(
```

```
  Sex = c("Male", "Female"),
  CVD = c("Detected", "Not detected")
)
# Complements: 54 - 5 = 49 and 288 - 4 = 284.
knitr::kable(addmargins(tab), caption = "Observed counts with marginal totals.")
```

Table 1: Observed counts with marginal totals.

	Detected	Not detected	Sum
Male	5	49	54
Female	4	284	288
Sum	9	333	342

```
stopifnot(sum(tab) == 342, all(rowSums(tab) == c(54, 288)),
          all(colSums(tab) == c(9, 333)))
```

Before plotting: which sex contributes more participants? Which contributes more detected cases? Are those two questions equivalent to comparing the proportion with detected CVD?

```
dat <- as.data.frame(as.table(tab))
dat$Sex <- factor(dat$Sex, levels = rownames(tab))
dat$CVD <- factor(dat$CVD, levels = colnames(tab))
dat$sex_total <- as.numeric(rowSums(tab)[as.character(dat$Sex)])
dat$status_total <- as.numeric(colSums(tab)[as.character(dat$CVD)])
dat$p_within_sex <- dat$Freq / dat$sex_total
dat$p_within_status <- dat$Freq / dat$status_total
sex_axis <- setNames(paste0(rownames(tab), "\n(n = ", rowSums(tab), ")"), rownames(tab))
status_axis <- setNames(paste0(colnames(tab), "\n(n = ", colSums(tab), ")"), colnames(tab))
```

Absolute frequencies: how many participants?

Question: how many participants in each sex category had CVD detected or not detected?

```
p_counts <- ggplot(dat, aes(Sex, Freq, fill = CVD)) +
  geom_col(position = position_stack(reverse = TRUE), width = .65, colour = "white") +
  scale_fill_manual(values = cvd_colours) +
  scale_x_discrete(labels = sex_axis) +
  scale_y_continuous(expand = expansion(mult = c(0, .05))) +
  labs(title = "Colour-vision status by sex",
       subtitle = "Counts: the two sex groups have different sample sizes",
       x = "Sex (as recorded)", y = "Number of participants", fill = "CVD") +
  theme(legend.position = "bottom")
p_counts
```

The sample contains many more females than males. The blue segments nevertheless represent more detected cases among males: five versus two. Counts correctly answer a question about **numbers of cases**. They do not directly answer a question about the fraction affected within each group.

Compare the four counts with equal-width bars

```
ggplot(dat, aes(Sex, Freq, fill = CVD)) +
  geom_col(position = position_dodge(width = .8, preserve = "single"), width = .7) +
  geom_text(aes(label = Freq), position = position_dodge(width = .8), vjust = -.4) +
```

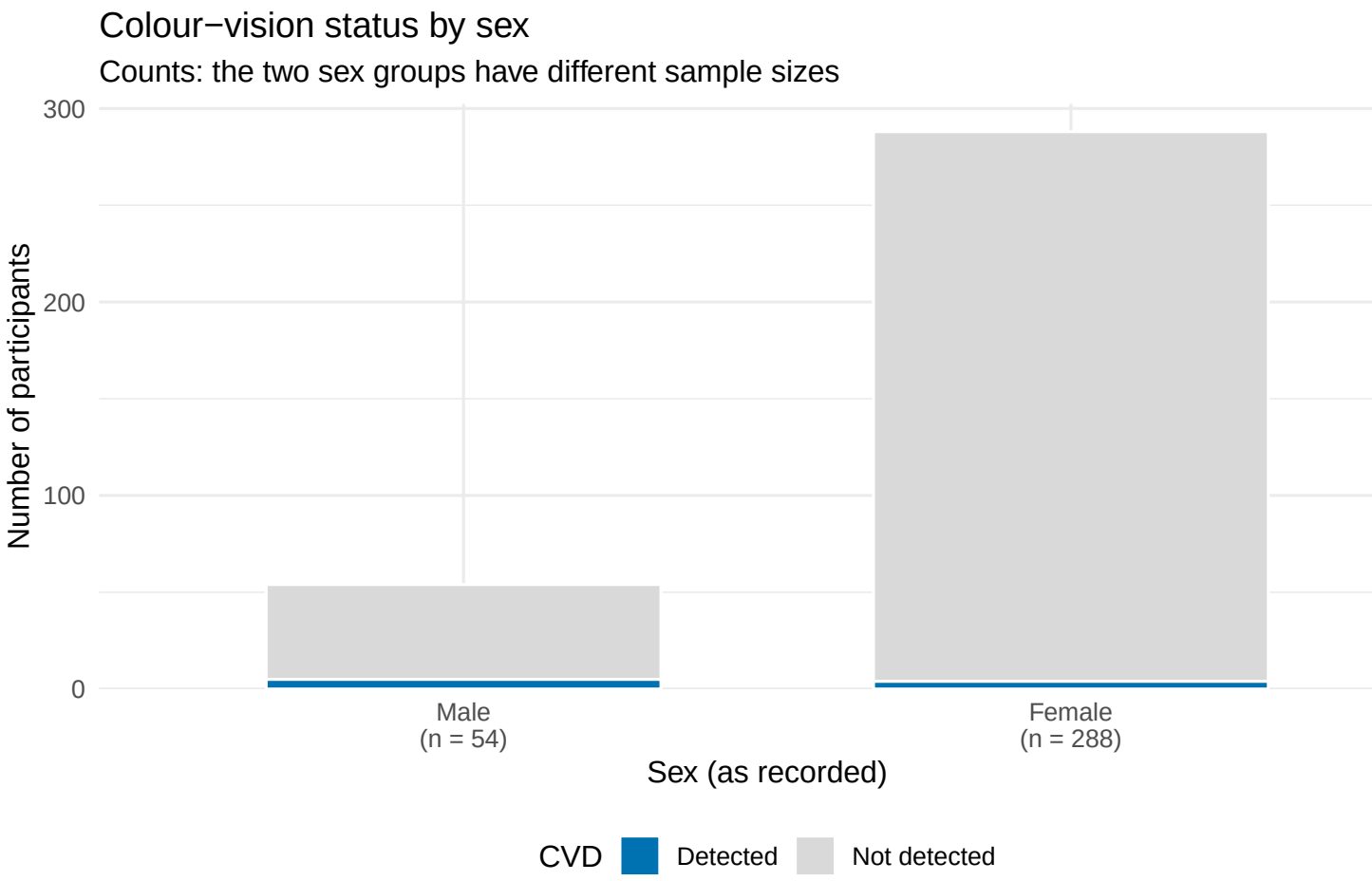
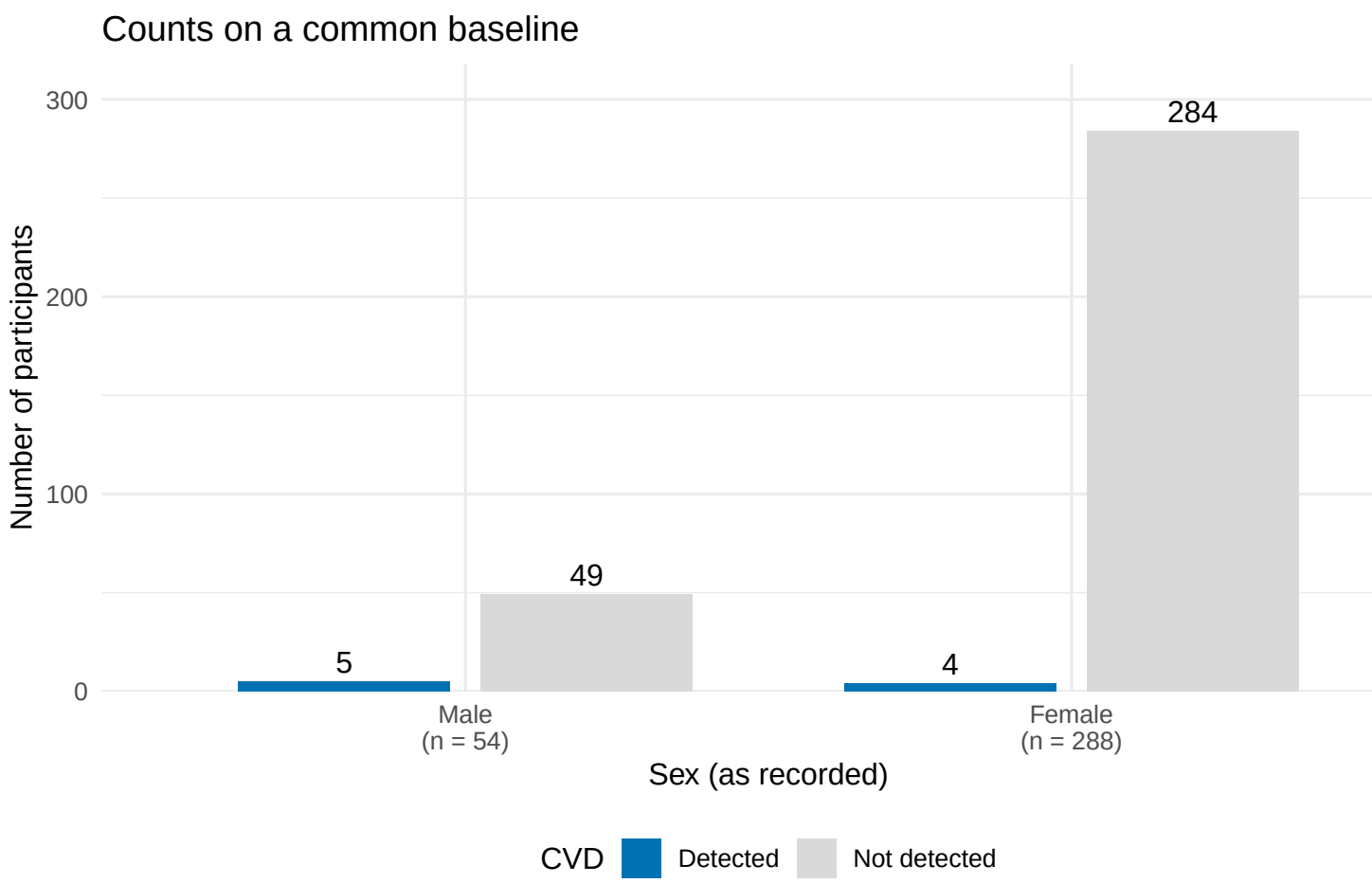


Figure 1: Stacked counts show both sample sizes and outcome counts. The detected segment starts at zero.

```
  scale_fill_manual(values = cvd_colours) +
  scale_x_discrete(labels = sex_axis) +
  scale_y_continuous(expand = expansion(mult = c(0, .12))) +
  labs(title = "Counts on a common baseline", x = "Sex (as recorded)",
       y = "Number of participants", fill = "CVD") +
  theme(legend.position = "bottom")
```



The side-by-side bars make each outcome count directly comparable. Their baseline remains zero, and the labels make the small counts visible.

Conditional proportions: colour-vision status given sex

Question: within each sex group, what proportion had CVD detected?

For a sex category s, the empirical conditional proportion is

$$\widehat{P}(\text{CVD} = \text{Detected} \mid \text{Sex} = s) = \frac{n_{s,\text{Detected}}}{n_{s,+}}.$$

Here the denominator is a **row total**, since rows correspond to sex.

```
within_sex <- prop.table(tab, margin = 1)
knitr::kable(100 * within_sex, digits = 2,
             caption = "Percentages within sex: each row sums to 100%.")
```

Table 2: Percentages within sex: each row sums to 100%.

	Detected	Not detected
Male	9.26	90.74
Female	1.39	98.61

```
stopifnot(all(abs(rowSums(within_sex) - 1) < 1e-12))
```

Bars normalised to 100%

```
p_within_sex <- ggplot(dat, aes(Sex, Freq, fill = CVD)) +
  geom_col(position = position_fill(reverse = TRUE), width = .65, colour = "white") +
  scale_fill_manual(values = cvd_colours) +
  scale_x_discrete(labels = sex_axis) +
  scale_y_continuous(labels = scales::label_percent(),
                     expand = expansion(mult = c(0, .02))) +
  labs(title = "Colour-vision status within each sex",
       subtitle = "Each bar represents 100% of its sex group",
       x = "Sex (as recorded)", y = "Proportion within sex", fill = "CVD") +
  theme(legend.position = "bottom")
p_within_sex
```

The detected proportions are **5/54 = 9.26% among males** and **4/288 = 1.39% among females**. Normalisation removes the difference in total bar heights; the labels retain the sample sizes.

Show only the proportion of interest

Because CVD is uncommon in this sample, the blue segments occupy a small part of the 100% bars. If the question concerns only detected CVD, show that proportion directly.

```
detected <- subset(dat, CVD == "Detected")
detected$label <- sprintf("%d/%d = %.2f%%", detected$Freq,
                          detected$sex_total, 100 * detected$p_within_sex)
p_detected <- ggplot(detected, aes(Sex, p_within_sex)) +
  geom_col(fill = cvd_colours["Detected"], width = .55) +
  geom_text(aes(label = label), vjust = -.6) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1),
                     limits = c(0, .12), expand = expansion(mult = c(0, 0))) +
  labs(title = "Detected CVD as a proportion of each sex group",
       x = "Sex (as recorded)", y = "Proportion with CVD detected") +
  theme(panel.grid.major.x = element_blank())
p_detected
```

This graph preserves the same two conditional proportions while making their comparison easier. It is a descriptive display, not a demonstration of statistical precision. There are only seven detected cases; population inference would require attention to sampling and uncertainty.

Reverse the conditioning: sex given colour-vision status

A different question: among participants with CVD detected, what proportion were male or female?

Now the denominator is **seven detected cases**, not all males or all females:

$$\widehat{P}(\text{Male} \mid \text{Detected}) = \frac{5}{9} = 55.56\%, \qquad \widehat{P}(\text{Detected} \mid \text{Male}) = \frac{5}{54} = 9.26\%.$$

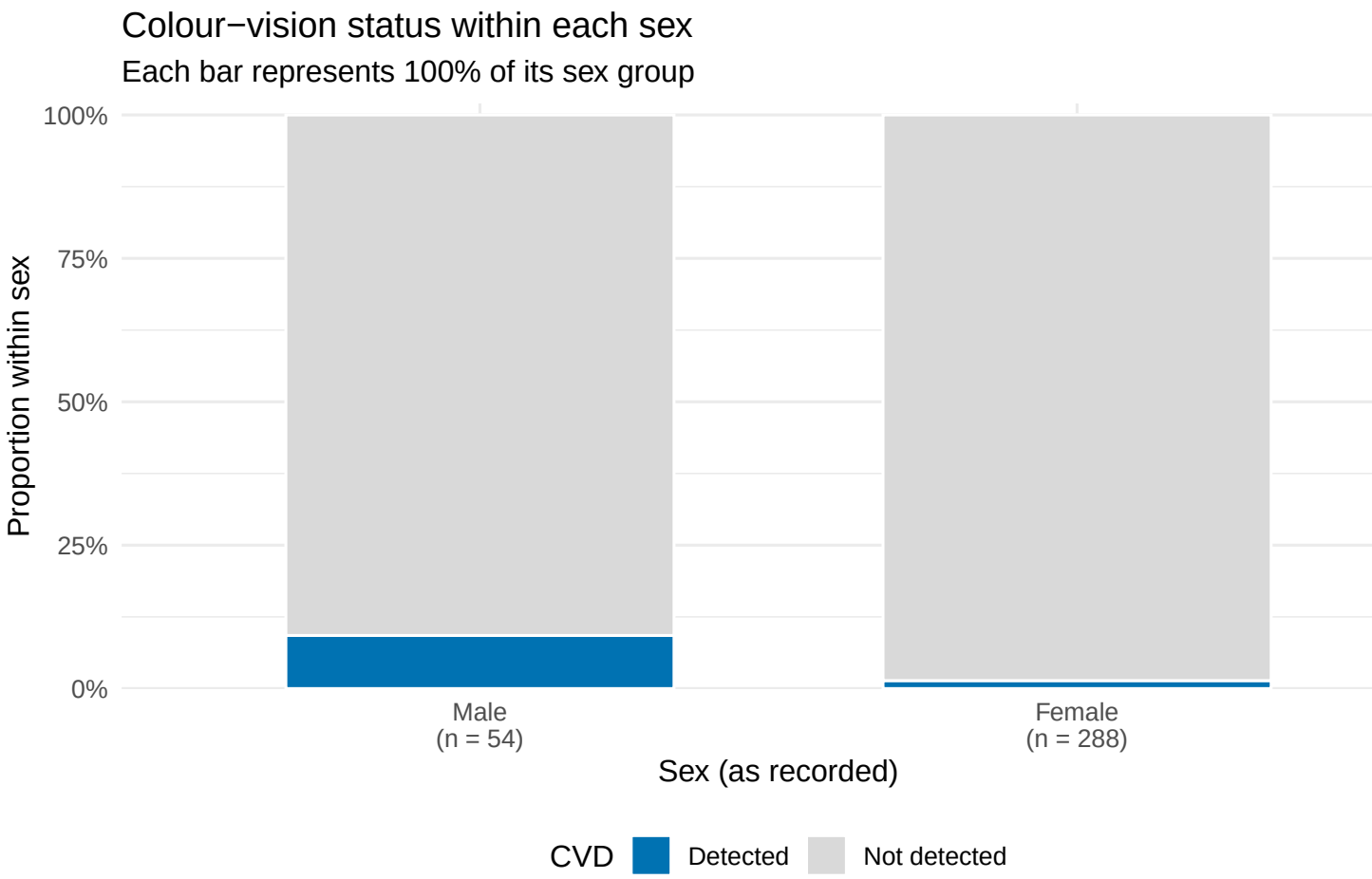


Figure 2: Each bar now has its own denominator: 54 males or 144 females. The blue segments share the zero baseline.

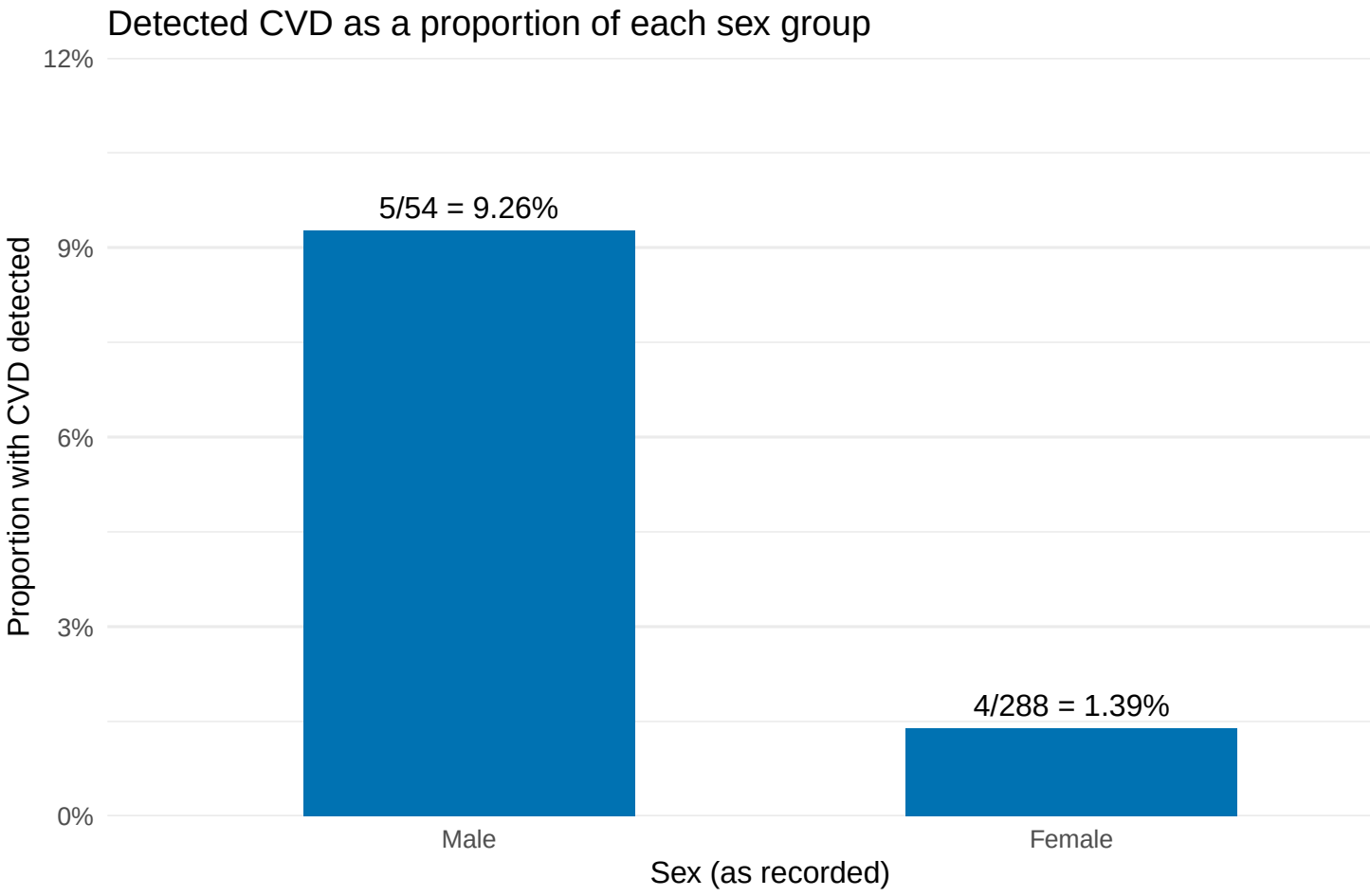


Figure 3: The axis starts at zero and focuses on detected CVD. Labels include both numerator and denominator.

The numerator is the same cell of the table. The denominator and interpretation are different.

```
within_status <- prop.table(tab, margin = 2)
knitr::kable(100 * within_status, digits = 2,
             caption = "Percentages within colour-vision status: each column sums to 100%.")
```

Table 3: Percentages within colour-vision status: each column sums to 100%.

	Detected	Not detected
Male	55.56	14.71
Female	44.44	85.29

```
stopifnot(all(abs(colSums(within_status) - 1) < 1e-12))
```

```
sex_colours <- c("Male" = "#4D4D4D", "Female" = "#BDBDBD")
p_reverse <- ggplot(dat, aes(CVD, p_within_status, fill = Sex)) +
  geom_col(position = position_stack(reverse = TRUE), width = .65, colour = "white") +
  geom_text(aes(label = sprintf("%s\n%.1f%%", Sex, 100 * p_within_status), colour = Sex),
            position = position_stack(vjust = .5, reverse = TRUE), show.legend = FALSE) +
  scale_colour_manual(values = c(Male = "white", Female = "black"), guide = "none") +
  scale_fill_manual(values = sex_colours) +
  scale_x_discrete(labels = status_axis) +
  scale_y_continuous(labels = scales::label_percent(),
                     expand = expansion(mult = c(0, .02))) +
  labs(title = "Sex composition within each colour-vision status",
       subtitle = "Each bar represents 100% of its status group",
       x = "CVD status", y = "Proportion within status", fill = "Sex (as recorded)") +
  theme(legend.position = "bottom")
p_reverse
```

The detected group is 71.43% male and 28.57% female; the not-detected group is 25.65% male and 74.35% female. These describe **group composition**. They are not sex-specific CVD proportions. Composition also depends on the sex distribution in the sample.

Compare the questions side by side

```
(p_counts + labs(title = "How many?", subtitle = NULL)) +
  (p_within_sex + labs(title = "What fraction within each sex?", subtitle = NULL)) +
  plot_layout(guides = "collect") & theme(legend.position = "bottom")
```

Quantity	Calculation	Question
Detected males	5	How many males had CVD detected?
Detected fraction among males	5/54	Among males, what fraction had CVD detected?
Male fraction among detected cases	5/7	Among detected cases, what fraction were male?
Joint sample proportion	5/198	What fraction of all participants were male and had CVD detected?

The appropriate graph depends on the question. Counts show numbers; conditional proportions

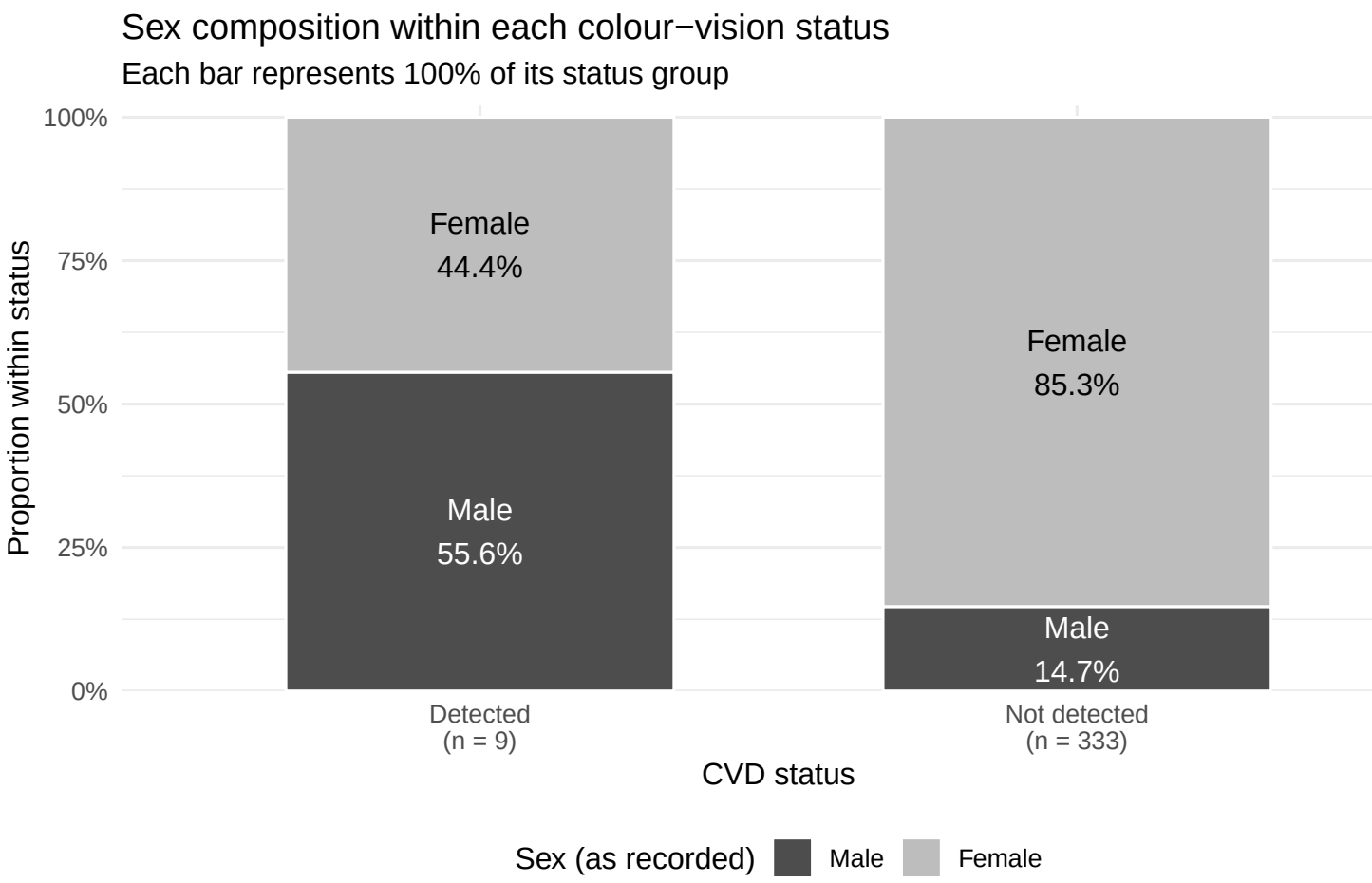


Figure 4: The denominator is now the number of participants in each colour-vision status category.

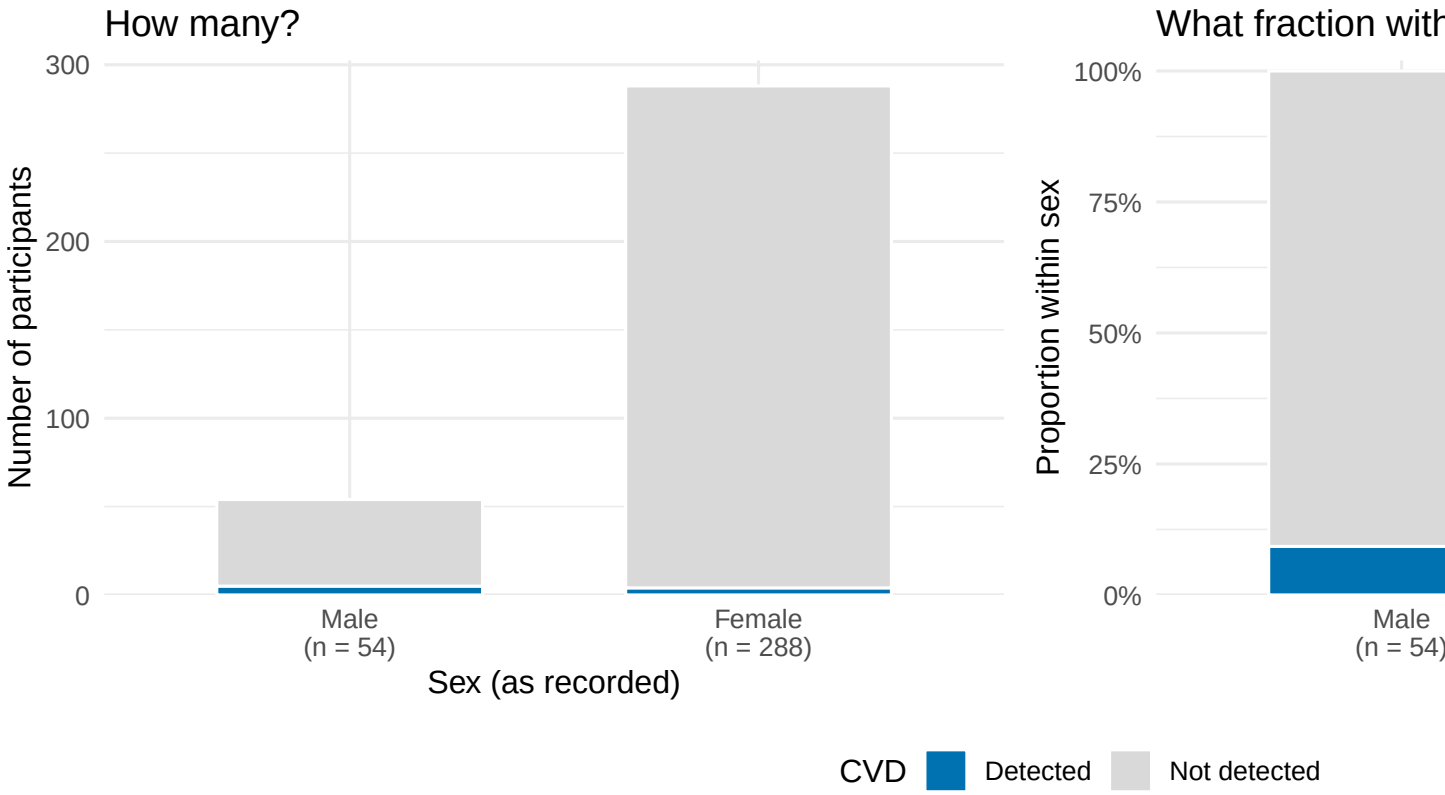


Figure 5: The same table supports counts and conditional proportions, but these answer different questions.

compare distributions within explicitly defined groups. Changing the conditioning variable changes the denominator.

Optional: a mosaic plot

A mosaic plot can encode group size and conditional distribution together. In the plot below, column widths represent the sex-group proportions; segment heights represent the CVD proportions within each sex. Rectangle areas therefore represent joint sample proportions.

```
mosaicplot(tab, color = unname(cvd_colours[colnames(tab)]),
           main = "Sex and colour-vision status",
           xlab = "Sex (as recorded)", ylab = "CVD status", las = 1)
```

The narrower male column reflects its smaller sample size. The small detected group makes some rectangles thin; the table and direct labels in earlier plots remain useful for exact reading. A mosaic plot provides more information at once but requires more explanation than aligned bars.

Reference and reproducibility

Ngente, Z., Jadav, B., Aafaque, S., Patil, A., Govindarajulu, R., Muthusamy, P., et al. (2021). *Prevalence of Color Vision Anomalies among Dental Professionals.* **Middle East African Journal of Ophthalmology**, 28, 169–173. Data: Table 2. Full text.

To render, install `rmarkdown`, `knitr`, `ggplot2`, `scales`, and `patchwork`, then run:

```
rmarkdown::render("conditional_barplots_colour_vision.Rmd")
```

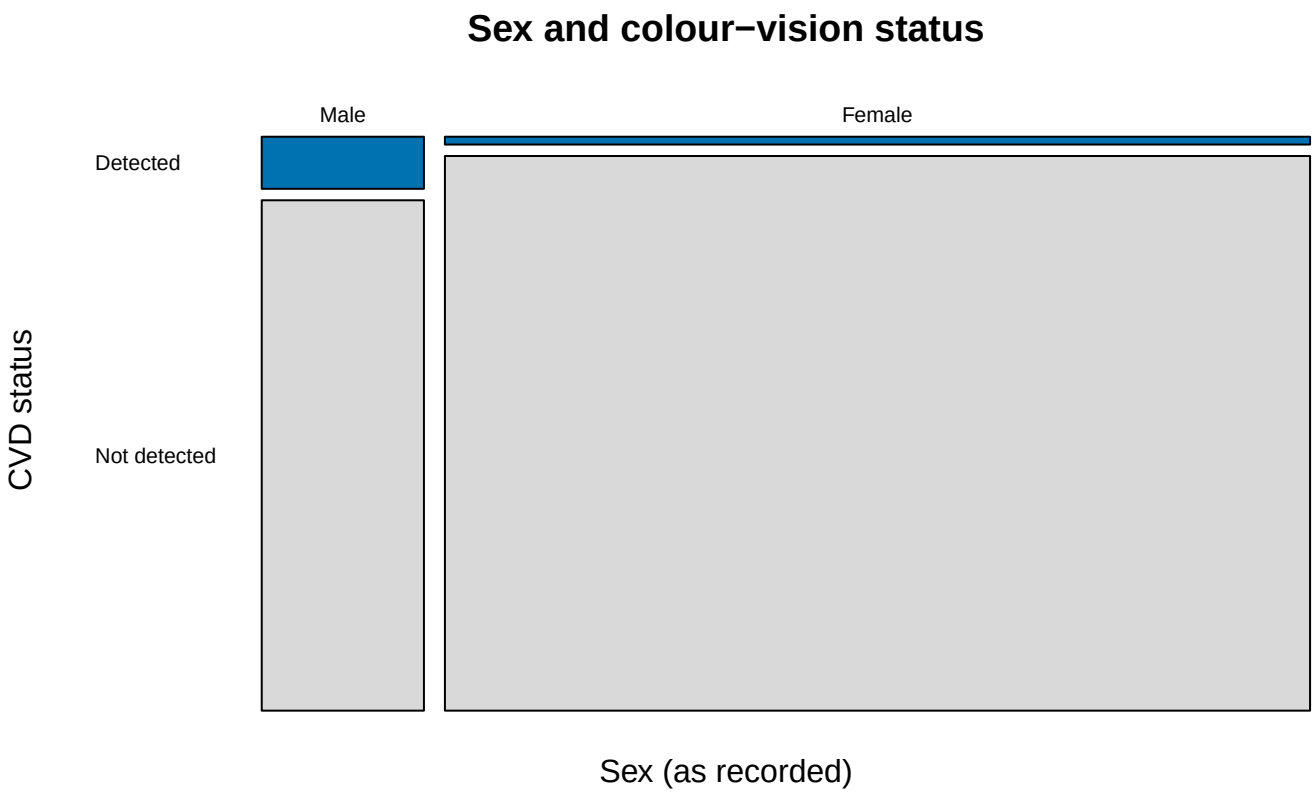


Figure 6: Mosaic plot: widths encode sex-group sizes; heights encode status proportions within sex; areas encode joint proportions.

The count arithmetic and document structure were checked during preparation. The R chunks were not executed in the preparation environment because R was unavailable.

Relationship between two count variables

Suppose we observe, for 443 students:

- `classes_attended`: number of classes attended, from 0 to 10;
- `assignments_submitted`: number of assignments submitted, from 0 to 12.

```
library(tidyverse)

set.seed(123)

students <- tibble(
  student_id = 1:300,
  classes_attended = rbinom(300, size = 10, prob = 0.65)
) |>
  mutate(
    p_submit = plogis(-1.5 + 0.30 * classes_attended),
    assignments_submitted = rbinom(300, size = 12, prob = p_submit)
  )

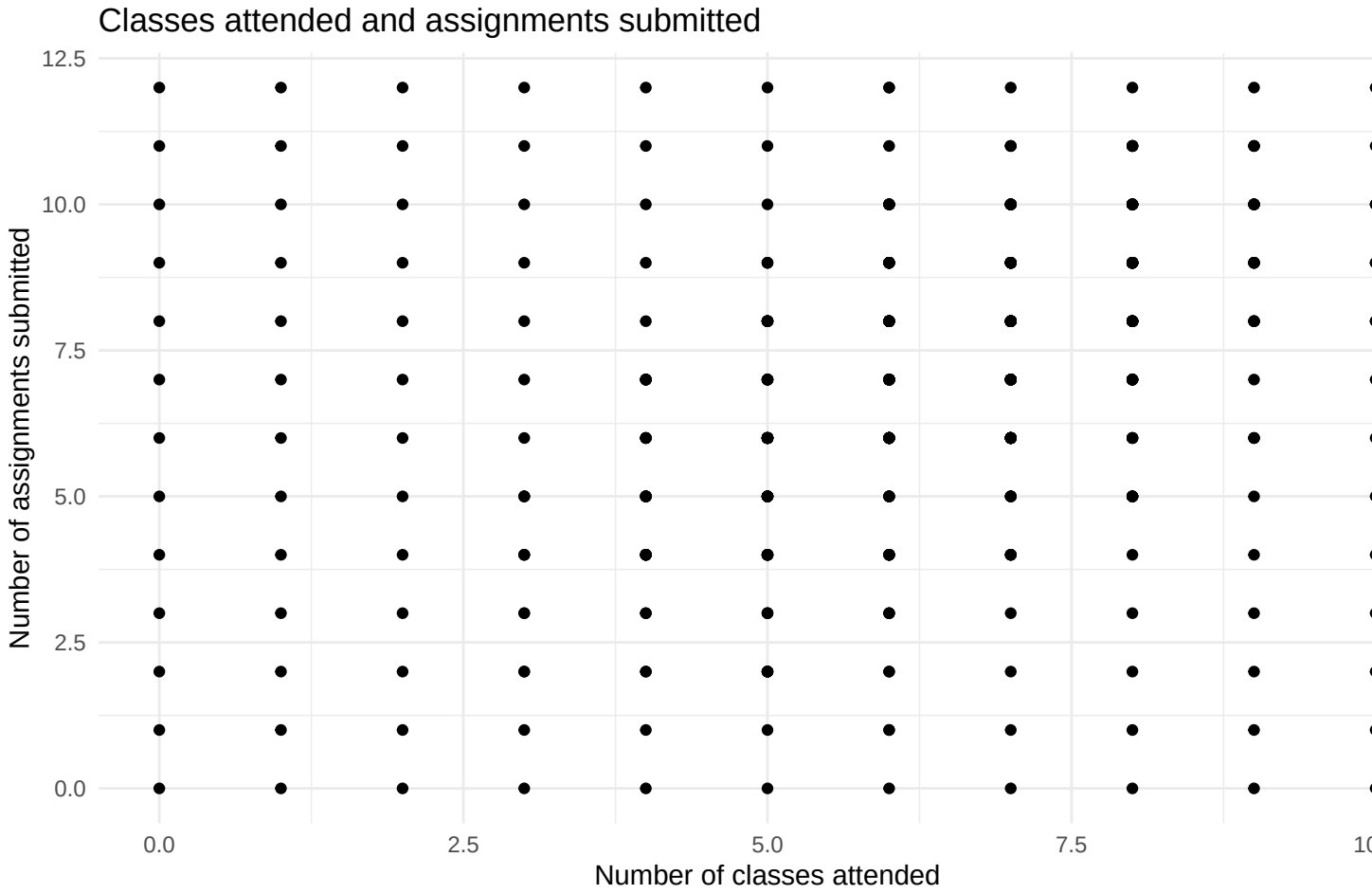
students$p_submit=NULL
eg=expand.grid(0:10,0:12)
colnames(eg)=colnames(students)[-1]
eg=data.frame(student_id=nrow(students)+1:nrow(eg),eg)
students=rbind(students,eg)

write.csv(file="./data/students.csv",students)
```

Why an ordinary scatter plot is inadequate

Both variables are discrete. Consequently, many students have exactly the same pair of values.

```
ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_point() +
  labs(
    title = "Classes attended and assignments submitted",
    x = "Number of classes attended",
    y = "Number of assignments submitted"
  ) +
  theme_minimal()
```

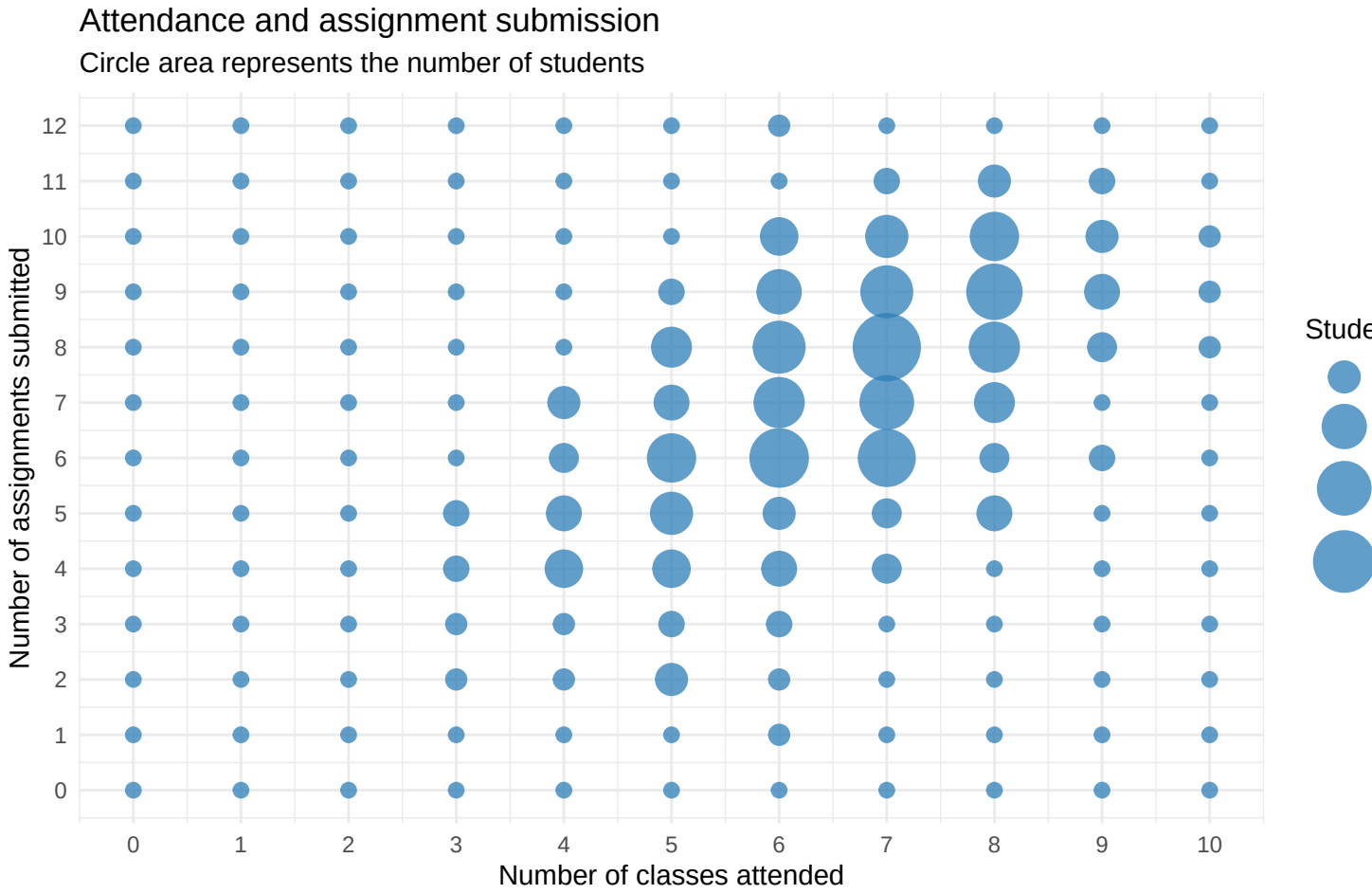


Several observations are drawn in precisely the same position. The graph therefore hides the frequency of the different combinations: a point may represent one student or many students.

Proposal 1: count plot

A count plot represents each observed combination with a circle whose area is proportional to its frequency.

```
ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_count(colour = "#2C7FB8", alpha = 0.75) +
  scale_size_area(max_size = 12) +
  scale_x_continuous(breaks = 0:10) +
  scale_y_continuous(breaks = 0:12) +
  labs(
    title = "Attendance and assignment submission",
    subtitle = "Circle area represents the number of students",
    x = "Number of classes attended",
    y = "Number of assignments submitted",
    size = "Students"
  ) +
  theme_minimal()
```



This graph preserves the observed integer values while making repeated combinations visible.

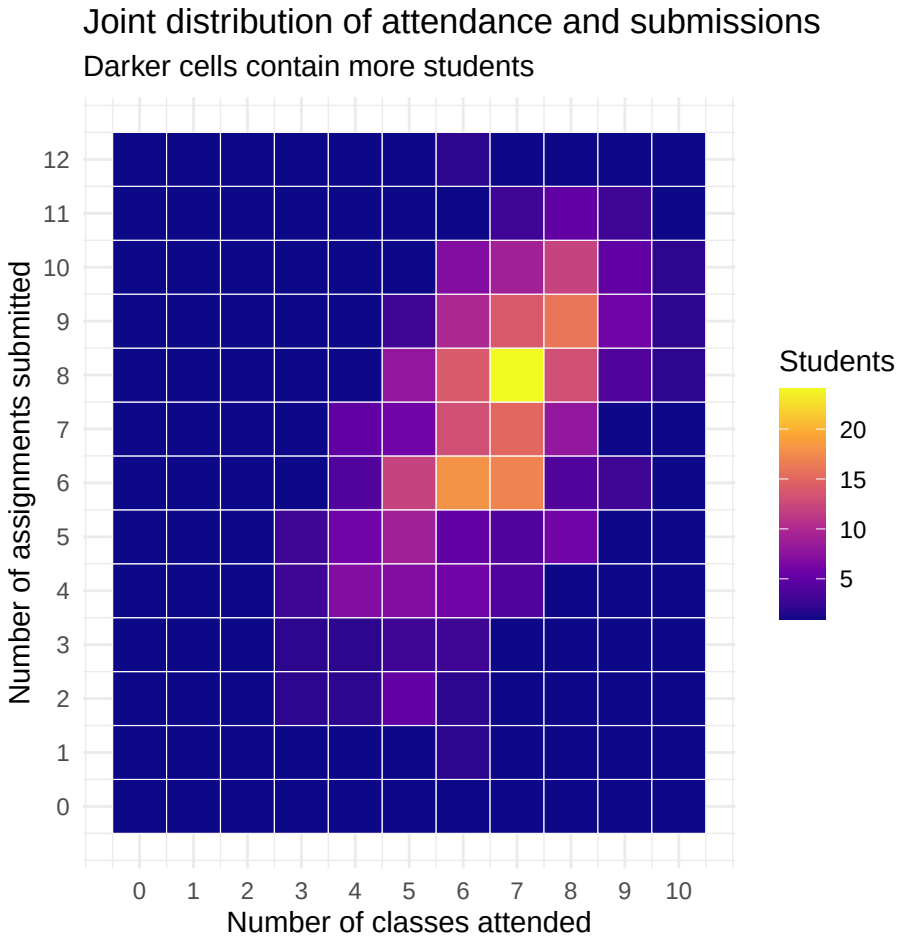
Proposal 2: joint-frequency heatmap

An alternative is to count the observations in each cell and represent frequency using colour.

```
joint_counts <- students |>
  count(classes_attended, assignments_submitted)

ggplot(
  joint_counts,
  aes(
    x = classes_attended,
    y = assignments_submitted,
    fill = n
  )
) +
  geom_tile(colour = "white") +
  scale_x_continuous(breaks = 0:10) +
  scale_y_continuous(breaks = 0:12) +
  scale_fill_viridis_c(option = "C") +
  coord_equal() +
  labs(
    title = "Joint distribution of attendance and submissions",
    subtitle = "Darker cells contain more students",
```

```
    x = "Number of classes attended",
    y = "Number of assignments submitted",
    fill = "Students"
  ) +
  theme_minimal()
```



The heatmap is particularly effective when there are many repeated combinations.

Proposal 3 Count plot with jitter

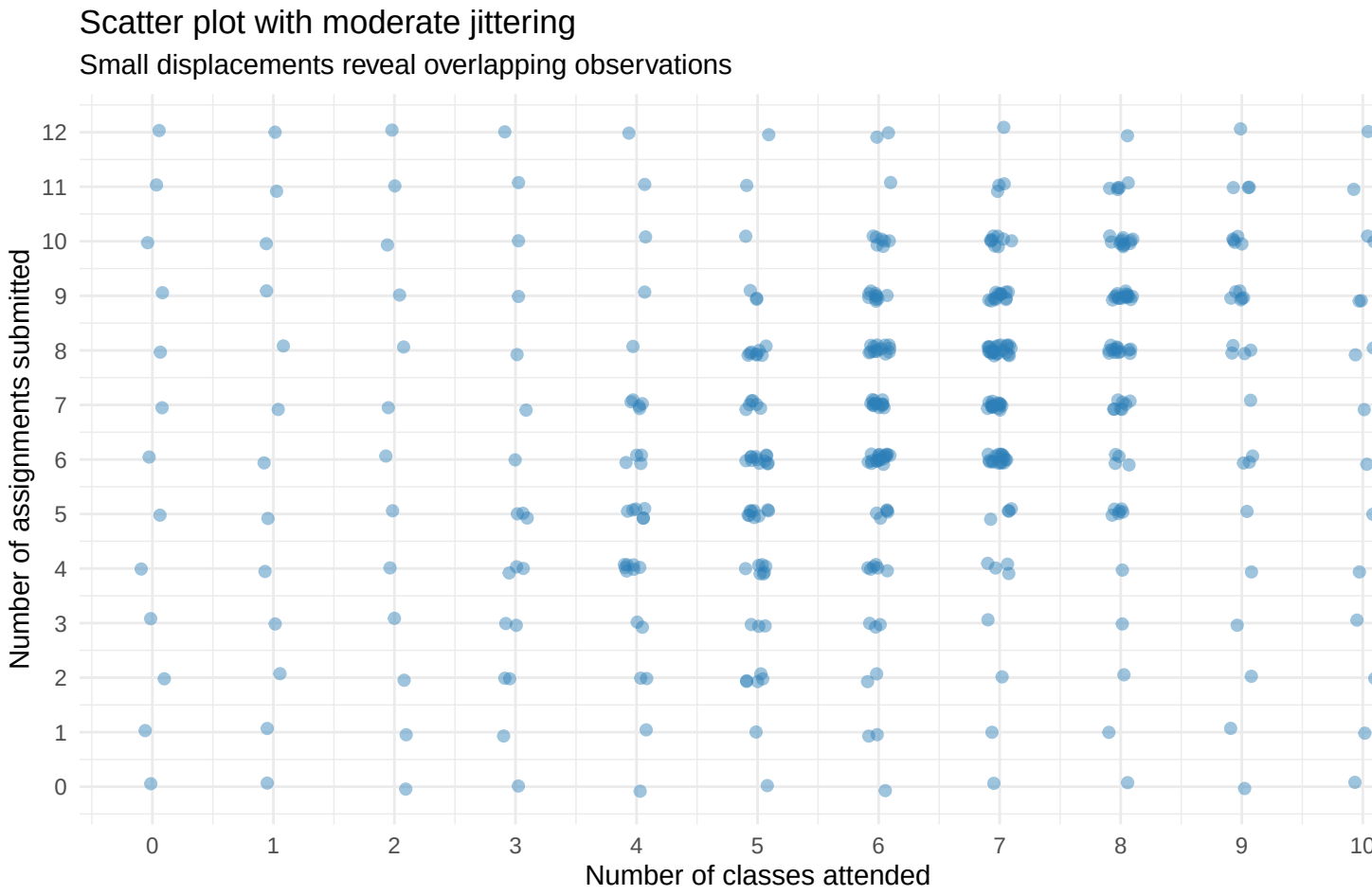
Scatter plot with a small amount of jitter

Jittering adds a small random displacement to overlapping points. The displacement makes individual observations more visible while preserving the discrete structure of the data.

```
set.seed(123)

ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_jitter(
    width = 0.10,
    height = 0.10,
    size = 1.8,
```

```
    alpha = 0.45,
    colour = "#2C7FB8"
  ) +
  scale_x_continuous(breaks = 0:10) +
  scale_y_continuous(breaks = 0:12) +
  labs(
    title = "Scatter plot with moderate jittering",
    subtitle = "Small displacements reveal overlapping observations",
    x = "Number of classes attended",
    y = "Number of assignments submitted"
  ) +
  theme_minimal()
```



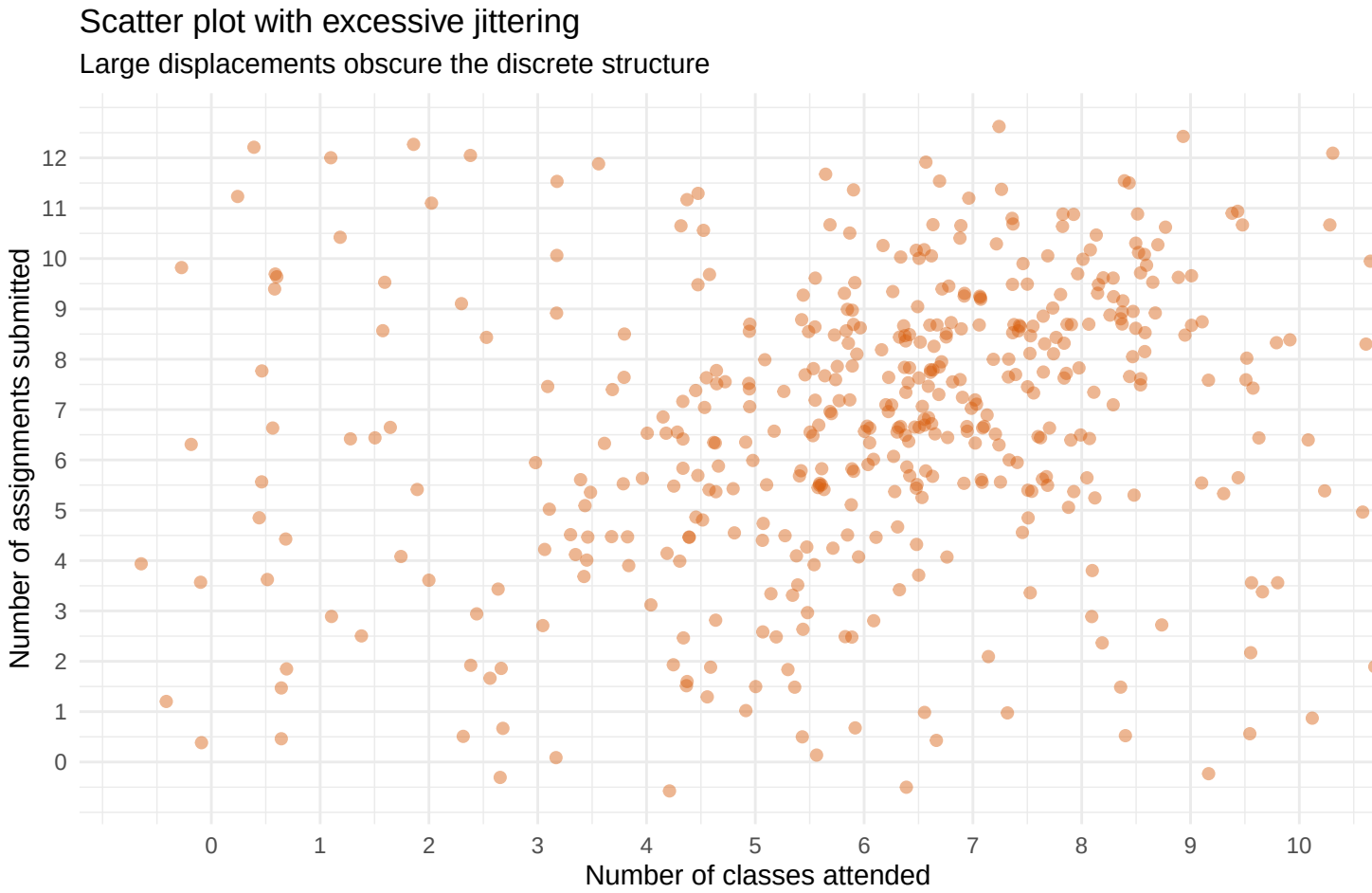
The original values are integers. The small horizontal and vertical displacements are graphical devices only: they do not represent measurement error or additional variation in the data.

Scatter plot with excessive jitter

```
set.seed(123)

ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
```

```
) +
  geom_jitter(
    width = 0.70,
    height = 0.70,
    size = 1.8,
    alpha = 0.45,
    colour = "#D95F0E"
  ) +
  scale_x_continuous(breaks = 0:10) +
  scale_y_continuous(breaks = 0:12) +
  labs(
    title = "Scatter plot with excessive jittering",
    subtitle = "Large displacements obscure the discrete structure",
    x = "Number of classes attended",
    y = "Number of assignments submitted"
  ) +
  theme_minimal()
```



Here the displacement is large relative to the distance between consecutive integer values. It creates the misleading impression that fractional counts were observed and makes neighbouring combinations difficult to distinguish.

Side-by-side comparison

```
library(patchwork)

p_count <- ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_count(colour = "#2C7FB8", alpha = 0.75) +
  scale_size_area(max_size = 10) +
  labs(
    title = "Count plot",
    size = "Students"
  ) +
  theme_minimal()

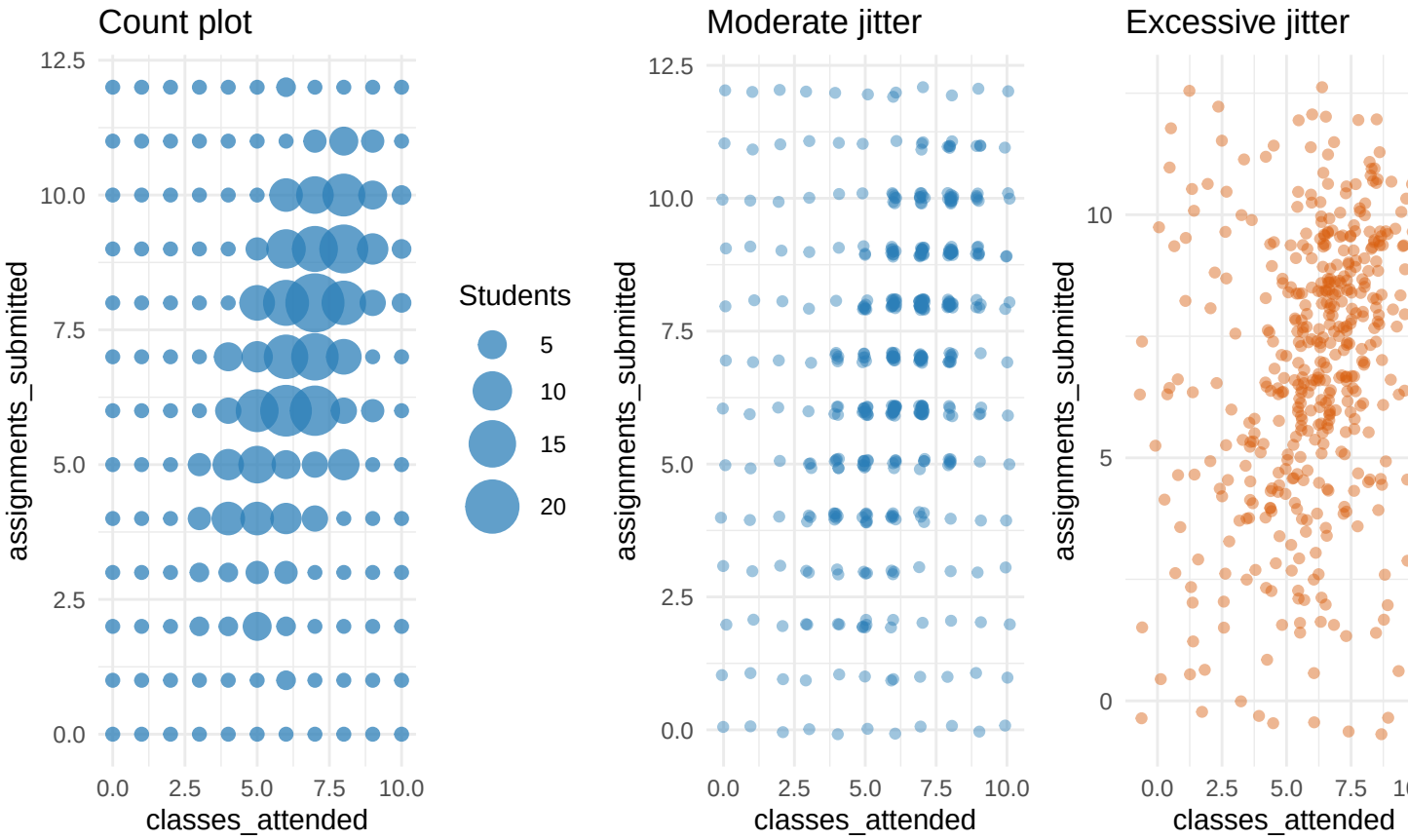
set.seed(123)

p_small <- ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_jitter(
    width = 0.10,
    height = 0.10,
    alpha = 0.45,
    colour = "#2C7FB8"
  ) +
  labs(title = "Moderate jitter") +
  theme_minimal()

set.seed(123)

p_large <- ggplot(
  students,
  aes(x = classes_attended, y = assignments_submitted)
) +
  geom_jitter(
    width = 0.70,
    height = 0.70,
    alpha = 0.45,
    colour = "#D95F0E"
  ) +
  labs(title = "Excessive jitter") +
  theme_minimal()

p_count + p_small + p_large +
  plot_annotation(
    title = "Three representations of two count variables"
  )
```



A useful question for the students is:

> At what point does jittering stop revealing overlap and start concealing the structure of the original data?

The key principle is that jitter should be small relative to the distance between possible values.

Suggested classroom question

> Both variables are counts. Why does the ordinary scatter plot provide an incomplete representation of their relationship? Which graphical element should represent the number of observations sharing the same pair of values?

Take-home message

> When two variables are discrete, observations often overlap. A count plot or joint-frequency heatmap displays both the relationship between the variables and the frequency of each observed combination.

Session Info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
```

```
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Italian_Italy.utf8  LC_CTYPE=Italian_Italy.utf8
## [3] LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
## [5] LC_TIME=Italian_Italy.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
##  [5] purrr_1.2.2     readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
##  [9] tidyverse_2.0.0 patchwork_1.3.2 ggplot2_4.0.3
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       compiler_4.6.1     tidyselect_1.2.1   tinytex_0.60
##  [5] scales_1.4.0       yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
##  [9] labeling_0.4.3     generics_0.1.4     knitr_1.51         tzdb_0.5.0
## [13] pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.3.0        stringi_1.8.9
## [17] xfun_0.60          S7_0.2.2           otel_0.2.0         viridisLite_0.4.3
## [21] timechange_0.4.0   cli_3.6.6          withr_3.0.3        magrittr_2.0.5
## [25] digest_0.6.39      grid_4.6.1         rstudioapi_0.19.0  hms_1.1.4
## [29] lifecycle_1.0.5    vctrs_0.7.3        evaluate_1.0.5     glue_1.8.1
## [33] farver_2.1.2       rmarkdown_2.31     tools_4.6.1        pkgconfig_2.0.3
## [37] htmltools_0.5.9
```

Note on AI assistance

These teaching materials were prepared with assistance from ChatGPT (OpenAI) in drafting explanations, developing examples, and generating R code. The author retains responsibility for the final content.